CL-ViME: Contrastive Learning and Vision Mixture of Experts for Encrypted Traffic Classification

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***Abstract*—Network traffic classification is essential for application identification and malicious behavior detection. However, the widespread use of encryption protocols hides payloads and reduces the availability of high-quality labeled data, both of which constrain the effectiveness of current models. To address these challenges, we propose CL-ViME, a self-supervised encrypted traffic classification framework that integrates Contrastive Learning and Vision Mixture of Experts. First, we design a packet-temporal matrix that preserves fine-grained packet headers and flow-level temporal structure. Second, we introduce a Vertical Vision Transformer-Mixture of Experts model to extract dual-view features through vertical patching and dynamic expert routing. Third, we develop a dual-granularity contrastive learning framework that aligns packet-level and flow-level representations via an MoE projector, followed by lightweight classifier-head fine-tuning. Experiments on three public datasets show that CL-ViME significantly outperforms state-of-the-art self-supervised and supervised baselines across accuracy, macro-precision, macro-recall, and macro-F1. It also demonstrates strong generalization and stability.**

***Index Terms*—Encrypted traffic classification, contrastive learning, mixture of experts, self-supervised learning.**

I. Introduction

NETWORK traffic classification [1], [2] has become a key technology for network management. Accurate classification helps understand network behaviors [3], identify applications [4], and detect malicious traffic [5], [6]. However, the widespread use of encryption technologies (such as TLS/HTTPS) renders most payload contents invisible. This obscurity complicates feature extraction and leads to performance bottlenecks in existing models [7], including reduced accuracy and limited generalization. Consequently, improving the accuracy of encrypted traffic classification has become a critical challenge.

Compared with machine learning (such as SVM [8], hierarchical clustering [9] and Random forest [10]), the powerful automatic feature extraction capability of deep learning provides a new direction for encrypted traffic classification. Deep learning models, such as convolutional neural network (CNN) [11], recurrent neural network (RNN) [12] and Transformer [13], can reduce the reliance on handcrafted features. However, they typically require large-scale, high-quality labeled data for training, yet obtaining accurate labels is costly. Moreover, enabling deep learning models to effectively recognize emerging traffic classes in dynamic network environments remains a significant challenge [14].

To reduce dependence on large-scale labeled data and improve generalization on unknown traffic, contrastive learning [15], [16] has gained attention for its self-supervised nature. Through designing suitable pretext tasks, it learns discriminative feature representations from large volumes of unlabeled data. It constructs positive and negative sample pairs, and then optimizes the loss function to pull positive pairs closer while pushing negative pairs apart. In this way, it autonomously captures category-level distinctions in network traffic. However, contrastive learning still faces several challenges in encrypted traffic classification [17], [18]: (1) Many models slice encrypted traffic into fixed-length byte sequences or gray-scale images. This approach loses fine-grained packet-level header semantics and disrupts flow-level temporal structures. (2) The priori design assumptions, such as kernel shapes, receptive field layouts and positional encodings, often mismatch the characteristics of encrypted traffic, leading to distorted feature extraction when directly transferred. (3) Most models build positive and negative pairs at only one granularity, either packet-level or flow-level. This single-level design limits their ability to exploit complementary information across granularities, making it hard to capture both microscopic protocol fingerprints and macroscopic behavioral patterns, and ultimately restricting generalization.

To address the above limitations, focusing on the challenges of feature representation, model adaptation and multi-granularity feature extraction, this paper proposes CL-ViME, a self-supervised encrypted traffic classification model built on Contrastive Learning and Vision Mixture of Experts. The main contributions are as follows:

- *Structured representation with dual-granularity fusion:* We first organize raw encrypted traffic into a

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two-dimensional matrix in a structured manner, and then fuse packet-level information (e.g., header fields) and flow-level information (e.g., temporal relationships) into a unified packet-temporal matrix. This representation preserves both fine-grained headers and macroscopic temporal patterns of encrypted traffic.

- *Vertical Vision Transformer-Mixture of Experts (Vertical ViT-MoE):* To address the anisotropic semantics of packet-temporal matrices, where horizontal and vertical axes encode intra-packet fields and inter-packet temporal sequences respectively, we replace the square patching of standard ViT with vertical segmentation. We further combine this strategy with a MoE to dynamically extract packet-level features from the horizontal dimension and flow-level features from the vertical dimension, producing dual-view representations for subsequent contrastive learning.
- *Dual-Granularity Contrastive Learning:* We first drive MoE projector to map the packet-level and flow-level representations into a shared latent space. We then use hybrid bidirectional contrastive loss to jointly optimize consistency and discrimination across granularities, enabling self-supervised alignment of packet- and flow-level features without labels and significantly improving robustness and generalization.
- *MoE Auxiliary Loss:* We introduce an MoE auxiliary loss that applies alignment-divergence regularization to expert routing. It enforces maximal routing divergence between different views of the same sample and maximal routing similarity across encoders for the same view. This constraint enables adaptive and robust alignment of packet-level and flow-level representations in a unified latent space.

The remainder of this paper is organized as follows: Section II reviews some related works. Section III detailed describes the CL-ViME model. Section IV provides the experimental setup. Section V presents and analyzes experimental results to validate the effectiveness of CL-ViME. Section VI summarizes the full paper and provides some future directions.

II. Related Works

This section reviews encrypted traffic classification techniques and contrastive learning models for traffic classification.

A. Encrypted Traffic Classification Techniques

With the develop of artificial intelligence, machine learning and deep learning techniques have been widely used in encrypted traffic classification.

1) Machine-Learning-Based Techniques: Machine learning-based techniques achieve automatic classification of different applications, protocols or malicious activities by analyzing the statistical characteristics and behavioral patterns of traffic packets. For example, a cost-sensitive SVM with active learning [8] improves generalization under class imbalance by selectively labeling minority classes and querying uncertain samples. However, it still relies on manual labeling. The ETC-PS method [10] models encrypted traffic through low-dimensional signature paths and achieves strong accuracy with high training efficiency. Nonetheless, its fixed feature dimensionality limits generalization in complex scenarios. Overall, traditional machine learning models depend on handcrafted features, while encrypted traffic conceals payload semantics and undermines the transferability of such features. Moreover, these models also struggle to capture the complex, high-dimensional patterns of encrypted traffic, resulting in limited robustness and adaptability in dynamic network environments.

2) Deep-Learning-Based Techniques: For the strong automatic feature extraction capability and the ability to model complex patterns, deep learning has become a mainstream approach for encrypted traffic classification. For example, FlowPic [11] converts encrypted traffic into a 2D histogram of arrival time and packet length,and applies CNNs for image-level classification. This design preserves privacy by avoiding payload inspection, but the ultra-high-resolution images contain sparse information because they rely on low-entropy features. Song et al. [12] proposed I^2RNN, an interpretable RNN that constructs fingerprint modules using LSTM. Each module learns the sequence patterns of one traffic class, enabling incremental learning without retraining the entire model. However, the number of modules grows linearly with the number of classes, increasing storage and inference cost. DigTraffic [13] adopts a message-level graph representation to capture client-server interactions using heterogeneous edge types. It encodes long and temporal flow sequences through a dual-channel network, and then employs a message-aware Graph Transformer that leverages node and edge information to model complex graph structures. However, its detection accuracy needs to be improved in scenarios with few labels. Deep learning reduces dependence on handcrafted features and demonstrates strong capability across byte-, packet-, and sequence-level representations. However, most models require large volumes of high-quality labeled data, which are difficult to obtain for encrypted traffic and thus limit generalization.

B. Contrastive Learning Models for Traffic Classification

Most mainstream traffic classification models rely on supervised learning, their performance is constrained by the scarcity of high-quality labeled data. Given the difficulty of labeling traffic and the continuous emergence of new traffic types, these models struggle to adapt quickly to unseen classes. In contrast, contrastive learning leverages large volumes of unlabeled traffic for representation learning, enabling faster adaptation to evolving network environments.

The FlowPic-based contrastive learning framework [15] learns the representations from unlabeled traffic, and groups flows within the same category. This model reduces reliance on large labeled datasets and improves performance on minority classes. The ContraMTD [16] combines CNN-based local feature extraction with a dual-edge attention GNN for global interactions. Its contrastive learning design supports cross-network training and enhances malicious traffic detection, but its generalization to broader encrypted traffic tasks remains limited. The pretraining framework CETP [17] learns generalized features from imbalanced encrypted traffic using contrastive learning and occlusion sequence modeling, and then uses fine-tuning to further mitigate class imbalance. However, its positive sample construction relies solely on random Dropout, which is often insufficient for low-redundancy encrypted traffic. Shen et al. [18] proposed a contrastive learning-based detector, it converts traffic into a semantic attribute matrix and pretrains on unlabeled data to capture normal behavior patterns. This design supports rapid adaptation to new attacks and improves robustness against obfuscation with limited labels. The multi-instance contrastive learning model MIETT [19] uses a two-level attention mechanism for packet- and flow-level features. It employs masked flow prediction, packet position prediction, and flow-level contrastive learning to improve traffic discrimination, but its accuracy still lags behind supervised models.

Researches in this area remain preliminary, and current models still trail state-of-the-art supervised models in classification accuracy. Existing contrastive tasks, typically adapted from computer vision, fail to capture the spatiotemporal structure and contextual dependencies inherent in network traffic. This study leverages the structured nature of encrypted traffic to derive refined packet- and flow-level representations and introduces a dual-granularity contrastive learning framework to strengthen classification capability. Experiments show that the proposed model achieves accuracy comparable to supervised deep learning models.

III. Methodology

This section describes the proposed CL-ViME, a self-supervised encrypted traffic classification model based on Contrastive Learning and Vision Mixture of Experts. It improves existing models by addressing the key challenges in feature representation, model adaptation and dual-granularity contrastive learning.

A. Framework of CL-ViME Model

The overall framework of the proposed self-supervised CL-ViME model is shown in Fig. 1. It comprises three components: a structured representation phase, a contrastive learning pretraining phase, and a fine-tuning phase.

Structured representation: This phase converts raw encrypted-traffic PCAP files into a packet–temporal matrix using structured padding. Matrix construction includes two steps: First, it horizontally pads packet protocols, flags and limited plaintext payloads. Second, it vertically embeds temporal patterns and interaction modes based on packet arrival order. This two-dimensional encoding jointly captures packet-level and flow-level semantics within a unified matrix structure. The resulting representation preserves fine-grained packet features and global flow dynamics, providing an effective input for subsequent contrastive learning.

Contrastive learning pretraining: This phase employs a self-supervised contrastive learning framework to exploit large-scale unlabeled traffic and enhance representation quality. Two data augmentation strategies, transposition and random masking, are first applied to construct discriminative positive and negative pairs, enabling the model to capture intrinsic characteristics of encrypted traffic. The Vertical ViT-MoE then serves as the backbone. It processes structured inputs through a vertically partitioned patch-embedding scheme, and uses a routing mechanism to dynamically select expert modules for feature extraction. This design jointly captures packet-level and flow-level semantics and routes heterogeneous features to the most suitable experts. With these operations, the pretrained model attains robust representation capability for downstream encrypted traffic classification.

Fine-tuning: This phase adopts a classification-head–only strategy. The pretrained Vertical ViT-MoE backbone is fully frozen, and only the task-specific classification head is updated. This approach verifies the generalization of the learned encrypted traffic representations while enabling efficient adaptation to downstream classification tasks with low computational overhead.

B. Structured Representation of Encrypted Traffic

In encrypted traffic classification, packet-level granularity treats each packet as the basic unit and captures fine-grained protocol fields and flags. However, it fails to model global behavioral patterns because it ignores inter-packet dependencies. Flow-level granularity aggregates bidirectional packets sharing the same 5-tuple (source/destination IP, source/destination port, and transport protocol) into a single unit to enable the extraction of temporal dynamics and interaction behaviors, albeit at the cost of losing detailed protocol and payload information. Existing representation methods either extract abstract features [20] and map them to pixel values [11], [21]-an approach that performs poorly on encrypted traffic due to weak feature extraction-or directly process raw traffic with simplistic single-granularity filling strategies, which disregard packet structure and lead to inefficient matrix usage and ambiguous semantic representations.

To address these challenges, we propose a structured feature representation method that constructs a packet-temporal matrix through dual-granularity fusion. The key idea is to directly populate raw encrypted traffic into a structured matrix, thereby integrating packet-level and flow-level information. At the packet level, the horizontal dimension encodes protocol fields, flags and limited payload content while preserving raw header semantics. At the flow level, the vertical dimension captures temporal dependencies and interaction patterns across packets. This dual-granularity representation preserves fine-grained details and simultaneously models global behavioral characteristics. The detailed process is as follows:

First, we reorganize packets in the PCAP file into bidirectional flows. Specifically, packets are first grouped into bidirectional sessions using the 5-tuple, and then sorted by timestamp to form interleaved, time-aligned round-trip flows of "source→destination" and "destination→source". Although payloads are encrypted, packet headers remain in plaintext and provide stable structured features. Their field distributions and semantics offer a reliable basis for encrypted traffic feature extraction.

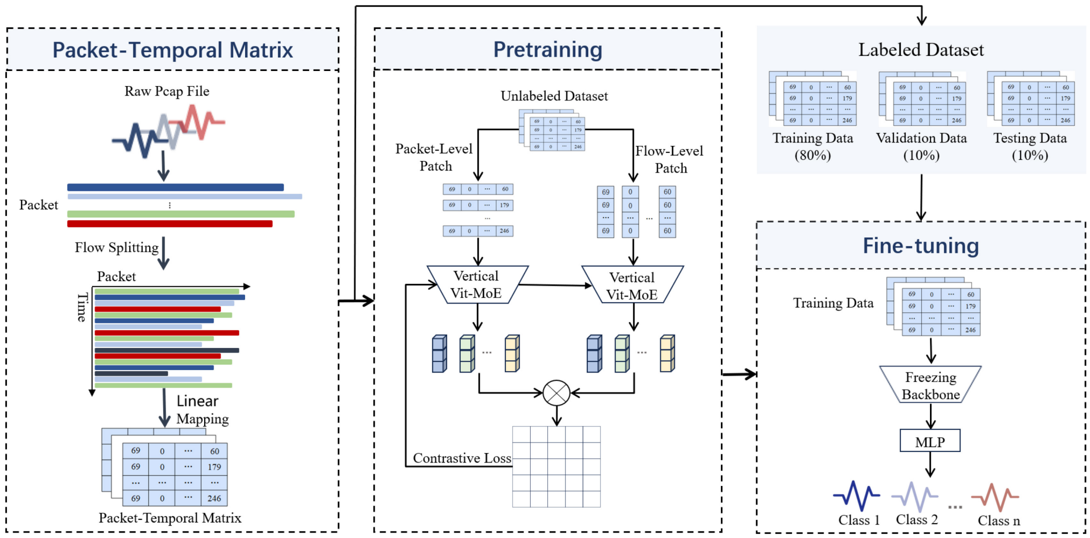


Fig. 1. The framework of CL-ViME model.

To mitigate interference from MAC and IP addresses and to prevent privacy leakage, we remove MAC addresses and anonymize source and destination IP addresses, retaining only the last-byte segment. This reduces the IP header to 13 bytes. The TCP header contains 20 fixed bytes and up to 40 optional bytes. Since optional fields rarely exceed 20 bytes, we allocate a uniform 40-byte space. Shorter headers are zero-padded, and longer ones are truncated to the first 40 bytes. In addition, although payloads are encrypted, some protocol-control fields following the TCP layer remain in plaintext and can be parsed. These fields are useful for downstream analysis, so we retain 7 bytes to capture essential communication behaviors while avoiding redundant or sensitive content. After these operations, each packet provides a fixed 60-byte header representation.

These 60 bytes are linearly mapped from 0-255 to decimal integers and placed into one row of a matrix, forming the horizontal spatial dimension. Each packet occupies a single row, and packets are stacked vertically by arrival order to form a 60×60 matrix. A bidirectional flow thus yields a sequence of matrices $S = [X_1, X_2, \ldots, X_n]$, where X_n denotes the n^{th} packet. This transformation preserves packet-header structure and embeds temporal patterns into the matrix layout. Horizontally, each row encodes the packet's internal fields and byte-level variations. Vertically, row-to-row differences capture temporal dependencies and interaction dynamics within the flow. By aligning intra-packet structure along rows and inter-packet sequences along columns, the method effectively exploits both spatial and temporal semantics of encrypted traffic.

C. Dual-Granularity Contrastive Learning Pretraining

Most labeled network traffic datasets (e.g., CICIOT2023 [22], CICIDS2018 [23]) are generated in virtual environments. Meanwhile, labeling real-world traffic is extremely difficult, which limits the applicability of supervised models due to the scarcity of reliable labels. A practical solution is to adopt self-supervised contrastive learning to classify encrypted traffic by exploiting unlabeled data. In contrastive learning, positive and negative pairs must be constructed, then the model minimizes the feature distance of positive pairs and maximizes that of negative pairs through the contrastive loss. The framework of proposed dual-granularity contrastive learning is shown in Fig. 2.

1) Construction of Positive and Negative Sample Pairs: In the computer vision domain, positive sample pairs for contrastive learning are typically generated using techniques such as random cropping. However, this method may destroy the integrity of structured features, such as those in packet-temporal matrices, which represent packet-level features horizontally and flow-level features vertically. To address this, we propose a novel method for constructing positive sample pairs based on row-column transposition. Specifically, as shown in Eq. (1), we first transpose the rows and columns of matrix to achieve symmetric transformation in the spatio-temporal dimension. Next, we introduce a random mask perturbation mechanism to the transposed matrix. This dual augmentation strategy preserves the semantic integrity of original encrypted traffic, enhances the feature robustness through structural reorganization and local masking.

$$x_1 = T_{mask}(X), x_2 = T_{mask}(T_{trans}(X)) \tag{1}$$

In Eq. (1), X is the transformed encrypted traffic matrix, $T_{mask}()$ denotes the random mask operation, $T_{trans}()$ denotes the transpose operation, (x_1, x_2) is the positive sample pairs.

Negative samples are typically drawn from other samples within the same training batch. Since each batch contains traffic matrices from different communication behaviors, protocol types or attack patterns, their semantic differences are substantial, making them suitable negative samples. Suppose a batch contains N independent samples $\{X_1, X_2, \ldots, X_N\}$. Each sample is augmented to produce two views (x_1, x_2), resulting in 2N samples in total. For any positive pair (x_1, x_2), the remaining 2(N-1) samples serve as negative samples. If X_i is the anchor, all samples other than $(x_{i,1}, x_{i,2})$ constitute

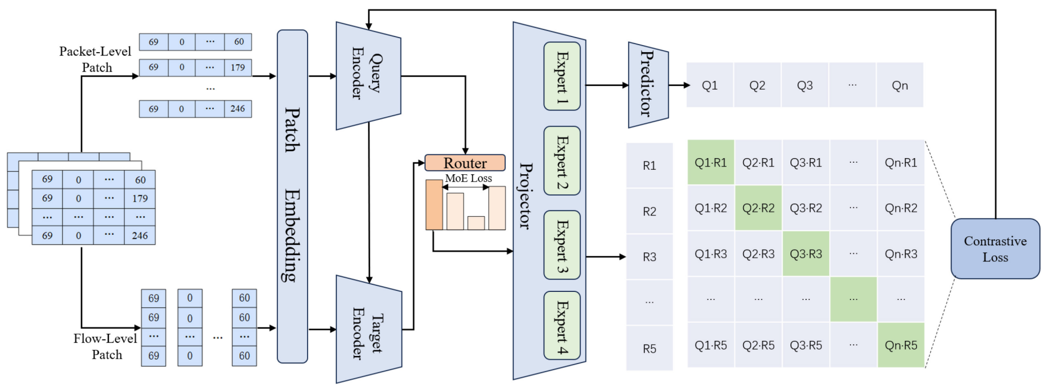


Fig. 2. The framework of dual-granularity contrastive learning.

its negative set, as shown in Eq. (2).

$$N_i = \left\{ \left(x_{j,1}, x_{j,2} \right) | j \in [1, N], j \neq i \right\} \tag{2}$$

In summary, the proposed method introduces a structured data augmentation strategy that preserves spatiotemporal integrity through matrix transposition and improves robustness through random masking. This approach avoids the structural distortion introduced by conventional cropping, it is well suited for feature representation scenarios of encrypted traffic with strict spatiotemporal dependencies.

2) Vertical ViT-MoE Model: Vision Transformer (ViT) [24] revolutionizes computer vision by splitting the images into uniform patches for sequence-based modeling. However, its isotropic 16×16 patching works well for natural images but not for encrypted traffic matrices. In those matrices, rows and columns carry different kinds of information: rows show protocol hierarchy, while columns show time dynamics.

To address this, we propose the Vertical Vision Transformer Mixture of Experts (Vertical ViT-MoE). It is a feature-oriented architecture with two key components: (1) Vertical Patching Strategy: Instead of using square patches, we segment inputs along the longitudinal axis. Given a positive sample pair (x_1, x_2), this approach extracts stream-level features from the original sample x_1 and packet-level features from the transposed sample x_2. (2) Dynamic Mixture-of-Experts (MoE): We integrate a dynamic MoE mechanism into the Transformer encoder. This allows specialized experts to capture the multi-granular semantic patterns that align with the inherent vertical structure of network traffic.

Specifically, we first design a parameterized vertical segmentation strategy for spatial feature decomposition. As shown in Eq. (3), it partitions the input $N * N$ matrix X into $N/2$ consecutive subregions along the horizontal dimension.

$$P = Partition(X) = \{P_i\}_{i=1}^{N/2}, P_i \in R^{N \times 2} \tag{3}$$

Each vertical bar region P_i is projected into a $2N$-dimensional embedding vector $e_i \in \mathbb{R}^D$ $(D = 2N)$ using a linear projection matrix W_p, as shown in Eq. (4). A learnable positional encoding vector $e_i^{\text{pos}} \in \mathbb{R}^D$ is added to preserve spatial order, and $\text{Flatten}(\cdot)$ denotes the vectorization operation.

$$e_i = \text{Flatten}(P_i)\, W_p + e_i^{pos} \tag{4}$$

A global [CLS] token is then prepended to the $N/2$ region embeddings to form the full input sequence T, as shown in Eq. (5).

$$T = \{e_i\}_{i=1}^{N/2} + [e_{CLS}] \tag{5}$$

Next, T is processed by 12 stacked Transformer encoder layers. Each layer contains:

(1) A multi-head self-attention mechanism [25]. It uses 8 attention heads, where each head linearly transforms the embedding vectors e_i in T through W_Q, W_K and W_V to generate the query (Q), key (K) and value (V) matrices, as shown in Eq. (6).

$$[Q_i, K_i, V_i] = e_i \left[W_Q, W_K, W_V \right] \tag{6}$$

The attention weight α_i for the i^{th} embedding is then computed via dot product, as shown in Eq. (7), where d_h denotes the dimension of K_i.

$$\alpha_i = \text{Attention}(Q_i, K_i, V_i) = \text{softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_h}} \right) V_i \tag{7}$$

(2) MoE Feed-Forward Network [26]: The standard two-layer MLP (typically expanding to $4D$ is replaced with a Mixture-of-Experts (MoE) layer. Each expert is a single linear transformation with input and output dimensions equal to the patch-embedding dimension D. A routing module computes the expert probabilities s_i using a learnable matrix $W_{\text{router}} \in \mathbb{R}^{D \times E}$, as shown in Eq. (8), where Gaussian noise $\epsilon \sim \mathcal{N}(0, 1)$ is added to improve routing diversity. Due to only the top-2 experts are activated per token, the theoretical computation is reduced to roughly 25% of the original MLP.

$$s_i = \text{softmax}(\alpha_i\, W_{router} + \epsilon) \tag{8}$$

Based on the routing probabilities s_i, only the top-k expert outputs g_j are preserved using a Top-k sparse activation mechanism, as shown in Eq. (9). Here, $s \in \mathbb{R}^E$ is the routing-score vector, E is the number of experts, $\text{TopK}(s, k)$ returns the selected expert indices, and the indicator function $\mathbb{I}(\cdot)$ ensures that all non-selected experts are suppressed.

$$g_j = \frac{\exp\big(s_j \cdot \mathbb{I}(j \in \text{TopK}(s, k))\big)}{\sum_{m \in \text{TopK}(s,k)} \exp(s_m)}, \quad \forall j \in \{1, 2, \dots, E\} \tag{9}$$

The MoE output is then computed as the weighted sum of the activated expert outputs, as shown in Eq. (10), where y_i denotes the enhanced feature vector for the i^{th} vertical bar region P_i, and E_j is the j^{th} expert.

$$y_i = \sum_{j=1}^{k} g_j \cdot E_j(x) \tag{10}$$

Finally, the [CLS] token aggregates the global contextual information to produce final high-dimensional representation.

3) Design of Loss Function: In contrastive learning, target encoder stabilizes feature representations of negative samples to improve training robustness and consistency. It shares the same architecture as the query encoder but updates its parameters via a momentum rule rather than backpropagation [27], as shown in Eq. (11), where θ_k and θ_q are the parameters of target and query encoders, respectively, and $m \approx 1$ is the momentum coefficient.

$$\theta_k = m \cdot \theta_k + (1 - m) \cdot \theta_q \tag{11}$$

Traditional contrastive learning often employs a nonlinear projector-typically an MLP-after the encoder to map features into an embedding space, emphasizing invariant semantic content [28]. The packet-level and flow-level perspectives of encrypted traffic belong to same segment but from different representational perspectives, requiring the projector to be able to understand and process these two heterogeneous representations separately and map them to a unified latent space. To unify this representation, we replace original single projectors in the query and target encoders with a shared MoE (Mixture-of-Experts) projector. This design enables parallel processing of encoder features through multiple expert networks, with a gating router dynamically selecting and combining experts per sample. It enhances model expressiveness and nonlinearity while preserving consistency between flow-level and packet-level features.

Specifically, dual-view positive pairs (x_1, x_2) are generated via data augmentation and encoded by query encoder (Vertical ViT-MoE) into features (z_1, z_2). They are then passed through unified MoE projector to yield expert representations (q_1, q_2) and routing weights (qr_1, qr_2), as shown in Eqs. (12) and (13), where W_{router} is a learnable weight matrix, E_j denotes the *j*-th expert, r_{ij} is the selection probability for expert *j* given feature z_i, and q_i is the resulting MoE-projected representation. A predictor network then maps (q_1, q_2) into the contrastive learning space, producing final query predictions (p_1, p_2). The target encoder follows the same pipeline to generate target representations (k_1, k_2) and routing weights (kr_1, kr_2) from (x_1, x_2). The target encoder follows the same pipeline to generate target representations (k_1, k_2) and routing weights (kr_1, kr_2) from (x_1, x_2).

$$r_i = \text{softmax}(z_i W_{\text{router}}) \tag{12}$$

$$q_i = \sum_{j=1}^{k} r_{ij} E_j(z_i) \tag{13}$$

Next, to make expert selection in the MoE module more discriminative and adaptive, we design an auxiliary loss L_{moe}. This loss strengthens model's adaptability and expressiveness by refining the expert assignment strategy. The loss consists of two components:

(1) Similarity minimization loss L_{moe_view} for dual-view routing weights: This loss drives the routing weights (qr_1, qr_2) of two augmented views (x_1, x_2) of the same sample to diverge. Each view is generated through distinct data augmentation. By encouraging inconsistent expert activations across views, the loss promotes diverse expert usage and improves representation robustness. The formulation is shown in Eq. (14), where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity.

$$L_{moe\_view} = \text{sim}(qr_1, qr_2) \tag{14}$$

(2) Similarity maximization loss L_{moe_enc} for hetero-encoder routing weights: This loss enforces high similarity between the routing weights (*qr*, *kr*) produced by the query encoder and the target (key) encoder for the same input view. It promotes consistent expert selection for identical semantic content, even when obtained through different encoding paths, thereby stabilizing expert activation. The formulation is shown in Eq. (15).

$$L_{moe\_enc} = -\frac{1}{2}(\text{sim}(qr_1, kr_1) + \text{sim}(qr_2, kr_2)) \tag{15}$$

Finally, L_{moe_view} and L_{moe_enc} are combined to form the MoE auxiliary loss L_{moe}, as shown in Eq. (16).

$$L_{moe} = L_{moe\_view} + L_{moe\_enc} \tag{16}$$

After imposing semantic consistency constraints on the expert routing mechanism through the MoE auxiliary loss (Eqs. (14)-(16)), the MoE projector maps heterogeneous packet-level and flow-level features into a unified latent space. This process yields more discriminative and aligned expert representations.

Next, we need to compute sample similarity to pull positive pairs closer and push negative pairs apart. The standard InfoNCE loss (shown in Eq. (17)) [29] is widely used to compute similarity, where q denotes the query features from the query encoder, k^+ represents the corresponding positive sample features from the target encoder, $\{k^-\}$ is a set of negative sample features, and τ is the temperature that controls the sharpness of the similarity distribution.

$$L_{InfoNCE}\left(q, k^+, \{k^-\}\right) = -\log \frac{\exp(\text{sim}(q, k^+)/\tau)}{\sum_{k^-} \exp(\text{sim}(q, k^-)/\tau)} \tag{17}$$

However, standard InfoNCE loss only considers unidirectional relations, i.e., the similarity between query features and positive sample features. It fails to leverage the semantic correspondence from dual-view augmented sample pairs. Consequently, the model may overfit to a specific viewpoint and neglect complementary information between different augmented views. This limitation reduces the generalization of learned representations. To address this, we propose a hybrid bidirectional contrastive loss based on InfoNCE, as shown in Eq. (18). It computes contrastive losses in both directions, leveraging mutual prediction between the two augmented views for more robust feature alignment.

$$L_{CL} = L_{InfoNCE}\left(p_1, k_2, \{k_2^-\}\right) + L_{InfoNCE}\left(p_2, k_1, \{k_1^-\}\right) \tag{18}$$

Algorithm 1 Dual-Granularity Contrastive Learning

Require: Batch_size n, Packet-temporal matrix X, Vertical ViT-MoE: θ, MoE Projector: M, Predictor P, Balancing parameter λ, Momentum m, Mask Augmentation T_{mask}, Transpose Augmentation T_{trans}

Ensure: Total loss L_{Total}

1: $V = []$ *// initialize to store sample pairs*
2: ***for*** *each sample* X_i *in* X ***do***
3: $\quad v_1 = T_{mask}(X_i)$ *// Flow-level view*
4: $\quad v_2 = T_{mask}(T_{trans}(X_i))$ *// Packet-level view*
5: $\quad V[i].append(v_1, v_2)$
6: ***end for***
7: θ_q, θ_k=θ.copy *// create query & target encoders*
8: ***for*** $i \in [1, 2n]$ ***do***
9: $\quad q_i,\ qr_i = M(\theta_q(V[i]))$
10: $\quad p_i = P(q_i)$
11: $\quad$ ***start*** *no-gradient operation*
12: $\quad\quad \theta_k = m \cdot \theta_k + (1 - m) \cdot \theta_q$ *// momentum update*
13: $\quad\quad k_i,\ kr_i = M(\theta_k(V[i]))$
14: $\quad$ ***end*** *no-gradient operation*
15: ***end for***
16: $L_{moe} = sim(qr_1, qr_2) - ((sim(qr_1, kr_1) + sim(qr_2, kr_2)) \cdot 0.5)$
17: $L_{CL} = L_{InfoNCE}(p_1, k_2, \{k_i, i \in [3, 2n]\})$
$\quad + L_{InfoNCE}(p_2, k_1, \{i \in [3, 2n]\})$
18: $L_{Total} = L_{CL} + \lambda \cdot L_{moe}$
19: ***return*** L_{Total}

Ultimately, the objective function is a weighted sum of these two losses, balanced by parameter λ (Eq. (19)). This formulation enables joint optimization of feature learning and expert routing mechanism.

$$L_{Total} = L_{CL} + \lambda L_{moe} \tag{19}$$

The process of proposed dual-granularity contrastive learning is shown in Algorithm 1.

This algorithm comprises three phases, each corresponding to a core contribution. **(1) Positive and negative sample construction (lines 1-6):** An empty view list V is initialized (line 1) to store augmented samples. For each raw matrix X_i, two augmented views are generated: a flow-level view v_1 by random masking (line 3), and a packet-level view v_2 by row-column transposition followed by masking (line 4). Both views are appended to V (line 5). These two views from the same X_i form positive pairs, and all others are treated as negative pairs. **(2) Query and target encoder construction (lines 7-15):** Each augmented view passes through Vertical ViT-MoE to obtain the query representation q_i and router logits qr_i (line 9). The predictor maps q_i to the contrastive space p_i (line 10). The target encoder freezes gradients and uses momentum (lines 11-14) to generate target representations k_i and logits kr_i. **(3) Loss computation (lines 16-19):** The MoE auxiliary loss L_{moe} minimizes cosine similarity between dual-view routers (qr_1, qr_2) while maximizing similarity between query and target routers (qr_i, kr_i) (line 16). The bidirectional InfoNCE loss L_{CL} is computed for positive pairs (qp_1, k_2) and (qp_2, k_1), treating all other views as negatives (line 17). Finally, L_{moe} and L_{CL} are combined to form L_{Total} (line 18).

The dual-view encoding and cross-level alignment design of CL-ViME makes its computational cost largely dominated by the Transformer blocks and MoE modules. Given L spatial patches and an embedding dimension D, the PatchEmbed layer has a complexity of $O(LD^2)$. The multi-head attention blocks add $O(36LD^2 + L^2D)$, while the MoE-based feed-forward layer incurs $O(24LD^2)$. Thus, the overall complexity of CL-ViME is $O(L^2D + LD^2)$, indicating strong dependence on both the embedding dimension and the number of patches.

D. Fine-Tuning of Encrypted Traffic Classification Model

After pretraining, the contrastive model acquires generalizable representations from large-scale unlabeled encrypted traffic, capturing deep spatiotemporal patterns across traffic types. To assess the transferability and quality of these representations, we apply head-only fine-tuning. The pretrained backbone is frozen to preserve its extraction capability, while only a lightweight classification head (e.g., a small MLP and a fully connected layer) is trained. The head is randomly initialized (e.g., Xavier initialization), and all backbone parameters remain fixed.

To mitigate severe class imbalance, which is common in encrypted traffic, we use a weighted cross-entropy loss. This loss increases penalties for minority-class errors and improves sensitivity to rare traffic types. We optimize the head with an efficient optimizer such as AdamW and a low learning rate, enabling fast convergence on limited labeled data.

This setup isolates the effect of pretrained features. Because the head serves only as a simple linear adapter, downstream classification performance directly reflects the quality and adaptability of the learned representations, providing a clean evaluation of the contrastive pretraining phase.

IV. Experimental Setup

To validate the effectiveness of proposed CL-ViME model, we compare it with five state-of-the-art models through extensive experiments.

A. Description of Encrypted Traffic Datasets

In the experiments, we use four encrypted traffic datasets to evaluate the performance of proposed CL-ViME model.

(1) MAWI [30]: It is maintained by WIDE project, capturing the encrypted traffic from a trans-Pacific backbone link between Japan and global Internet. It captures snapshots at fixed 15-minute interval using tcpdump and stores raw packets in PCAP format. This traffic has no manual labels, offering multi-scenario and prior-free characteristics suitable for self-supervised pretraining. We select the traffic from Jan. 1 to Jan. 5, 2025 for pretraining.

(2) ISCX-VPN [31]: It is released by the Canadian Institute for Cybersecurity, containing raw PCAP traffic for six VPN-encrypted application categories. It is widely used for evaluating encrypted traffic classification models.

(3) CICIoT [22]: It is designed for IoT environments, containing normal behaviors and multiple attack types and benign IoT traffic. It contains both encrypted and unencrypted flows, covering benign and malicious communication channels.

TABLE I
DATASET STATISTICS

Dataset	Classes	Samples per Class
ISCX-VPN	VoIP, Streaming, P2P, File, Chat, Email	40111, 25473, 6975, 6315, 1380, 409
CICIOT2023	Benign, DDoS, DoS, Mirai, Recon, Spoofing, Brute_Force	46254, 26458, 24540, 21950, 9555, 3544, 3398
CTU-Malware	Zeus, Normal, Shifu, Trickbot, Ursnif, HTBot, Miuref, Dridex, Emotet, Locky, Upatre	60105, 34912, 17116, 11387, 11105, 7645, 4904, 2866, 647, 47, 17

(4) CTU-Malware [32]: It is built for long-term malware traffic analysis, containing various malware families alongside normal traffic. It contains encrypted and unencrypted flows, and simulates realistic network environments with detailed attack records.

Table I summarizes the class composition and sample distribution of three labeled datasets used in the fine-tuning and evaluation stages. Due to MAWI dataset only used for self-supervised pretraining, it is not included in this table. Note that all datasets exhibit significant class imbalance—a common characteristic in real-world encrypted traffic—our experimental setup preserves it to reflect practical deployment scenarios.

During pretraining, the model uses only unlabeled encrypted traffic from MAWI for self-supervised representation learning, allowing a comprehensive evaluation of transferability and generalization. In the fine-tuning stage, the model is trained on three labeled datasets: ISCX-VPN, CICIOT2023 and CTU-Malware. Each dataset is stratified and randomly split into training, validation and testing datasets using an 8:1:1 ratio, preserving the original class imbalance and reducing sampling bias. This yields a stable and reliable evaluation environment for encrypted traffic classification.

B. Baselines

To evaluate CL-ViME, we compare it with six representative baselines.

(1) CETP [17]: A semi-supervised model using contrastive pretraining on multi-granularity token sequences. It performs fine-tuning with pseudo-label iteration and dynamic loss weighting for imbalanced traffic.

(2) SmartDetector [18]: A contrastive learning-based model embedding traffic sequences into context-aware vectors. It simulates obfuscation via data augmentation and pretrains the encoder on unlabeled traffic for robust detection.

(3) MIETT [19]: A multi-instance learning model treating packets as instances and flows as bags. It applies two-level attention to capture token- and packet-level relations and uses packet position prediction and flow contrastive learning for pretraining.

(4) MTC-MAE [15]: A masked autoencoder model converting traffic into gray-scale images for pretraining. It adopts fine-tuning with few labels to achieve high-precision classification.

(5) CL-FlowPic [33]: A self-supervised model using mini-FlowPic histograms. It employs SimCLR-based representation learning with RTT scaling, time shifting, and packet loss augmentations, clustering traffic from the same application in latent space.

(6) YaTC [34]: A masked autoencoder-based traffic transformer with multi-level flow representation. It encodes raw traffic into a hierarchical matrix and applies packet- and flow-level attention for efficient feature extraction. Subsequently, it is pretrained on unlabeled data via the MAE paradigm and fine-tuned on limited labeled samples for classification.

C. Evaluation Metrics

To evaluate the CL-ViME model, we use standard multi-class classification metrics. For a k-class problem, TP_i, FP_i, and FN_i represent the true positives, false positives, and false negatives for class i ($i = 1, \ldots, k$), and N denotes the total number of samples.

(1) Accuracy (Acc): It represents the percentage of correctly classified network traffic samples among all samples. It is calculated by Eq. (20).

$$\text{Accuracy} = \frac{\sum_{i=1}^{k} TP_i}{N} \tag{20}$$

(2) Macro-Precision (Ma-Pre): It measures the classification correctness of the model, computed as the unweighted mean of precision across all classes, as shown in Eq. (21).

$$\text{Macro} - \text{Precision} = \frac{1}{k}\sum_{i=1}^{k}\frac{TP_i}{TP_i + FP_i} \tag{21}$$

(3) Macro-Recall (Ma-Rec): It is the arithmetic mean of recall across all categories, giving equal weight to each class—particularly important under class imbalance. It is calculated by Eq. (22).

$$\text{Macro} - \text{Recall} = \frac{1}{k}\sum_{i=1}^{k}\frac{TP_i}{TP_i + FN_i} \tag{22}$$

(4) Macro-F1 (Ma-F1): It provides a harmonic balance between precision and recall at the class level, then averages uniformly across classes, as given in Eq. (23).

$$\text{Macro} - \text{F1} = \frac{1}{k}\sum_{i=1}^{k}\frac{2\,TP_i}{2\,TP_i + FP_i + FN_i} \tag{23}$$

D. Set of Parameters

In the pretraining phase, we train for 300 epochs with a learning rate of 3e-4, a batch size of 2048, and a momentum coefficient *m* of 0.99.

In the fine-tuning phase, we train for 50 epochs with a learning rate of 1e-2, a batch size of 512, and an equilibrium parameter λ of 0.9.

For both phases, the embedding dimension is 120, the number of ViT layers is 12, and the optimizer is AdamW. All experiments are conducted on a server using an NVIDIA A40 GPU.

TABLE II
CLASSIFICATION PERFORMANCE UNDER DIFFERENT FEATURE REPRESENTATIONS (%)

Dataset		ISCX-VPN				CICIOT2023				CTU-Malware			
Model	Format	Acc	Ma-Pre	Ma-Rec	Ma-F1	Acc	Ma-Pre	Ma-Rec	Ma-F1	Acc	Ma-Pre	Ma-Rec	Ma-F1
CNN	Feature Sequence	89.65±0.49	78.64±0.86	90.42±1.00	<u>83.26±0.48</u>	70.16±1.13	64.31±0.69	67.34±0.47	59.76±0.77	89.79±1.29	79.67±1.18	80.76±1.11	79.52±1.09
	Gray-scale Image	<u>93.14±0.42</u>	<u>86.89±0.78</u>	<u>85.54±0.79</u>	70.69±1.15	<u>86.61±0.92</u>	<u>76.99±1.30</u>	<u>66.62±0.83</u>	<u>69.77±0.91</u>	<u>94.69±0.57</u>	<u>89.61±0.43</u>	<u>87.01±0.67</u>	<u>87.79±0.62</u>
	Packet-Temporal Matrix	**97.93±0.22**	**90.20±1.21**	**95.56±0.76**	**92.63±0.92**	**90.72±0.16**	**87.44±0.70**	**79.57±0.40**	**82.75±0.43**	**98.28±0.15**	**93.21±1.54**	**92.81±0.86**	**92.80±0.95**
VGG	Feature Sequence	90.16±0.56	82.30±1.08	85.99±1.87	82.12±1.26	79.34±0.30	<u>69.09±0.56</u>	60.58±0.29	61.80±0.23	<u>93.06±0.27</u>	75.89±0.23	76.62±0.24	76.14±0.24
	Gray-scale Image	<u>92.31±0.19</u>	<u>84.66±0.60</u>	<u>89.32±0.44</u>	<u>86.76±0.25</u>	<u>81.72±0.73</u>	67.39±0.92	<u>69.36±0.51</u>	<u>67.10±0.68</u>	88.90±0.45	<u>79.88±0.57</u>	<u>83.22±0.88</u>	<u>80.38±0.42</u>
	Packet-Temporal Matrix	**97.92±0.20**	**86.01±1.29**	**97.85±0.34**	**90.49±0.98**	**84.26±0.67**	**74.17±1.82**	**73.48±1.10**	**72.96±0.96**	**97.15±0.20**	**83.21±0.85**	**93.57±1.87**	**85.94±0.79**
LSTM	Feature Sequence	89.82±0.21	78.02±0.55	88.94±1.77	82.57±0.88	70.97±0.30	63.58±0.50	66.26±0.67	59.77±0.34	<u>92.99±0.23</u>	83.21±0.39	83.34±0.23	83.05±0.34
	Gray-scale Images	<u>94.02±0.62</u>	<u>85.30±2.04</u>	<u>92.71±1.17</u>	<u>88.25±1.60</u>	<u>86.21±0.73</u>	<u>70.43±0.68</u>	<u>72.73±0.88</u>	<u>69.30±0.66</u>	91.72±0.82	<u>83.54±1.41</u>	<u>86.01±0.83</u>	<u>83.48±1.27</u>
	Packet-Temporal Matrix	**97.94±0.19**	**85.41±0.61**	**98.08±0.34**	**90.04±0.52**	**92.34±0.16**	**83.17±0.27**	**90.67±0.56**	**85.84±0.27**	**98.63±0.20**	**85.33±0.83**	**96.82±0.71**	**87.89±1.06**
RNN	Feature Sequence	91.57±0.14	<u>89.07±0.62</u>	77.60±0.44	79.28±0.56	73.55±0.14	67.59±0.68	56.77±0.28	57.82±0.31	93.80±0.13	<u>85.74±0.18</u>	79.02±0.79	80.03±1.14
	Gray-scale Image	<u>92.95±0.53</u>	81.15±1.36	<u>80.22±1.44</u>	<u>80.49±1.39</u>	<u>90.62±0.72</u>	<u>74.03±1.88</u>	<u>66.04±1.56</u>	<u>67.24±1.58</u>	<u>94.32±0.69</u>	85.03±1.05	<u>83.16±1.30</u>	<u>83.88±1.22</u>
	Packet-Temporal Matrix	**97.77±0.19**	**93.04±1.03**	**80.96±1.91**	**84.37±2.06**	**92.23±0.19**	**91.14±0.56**	**83.76±0.66**	**86.86±0.44**	**98.97±0.31**	**98.74±0.33**	**93.63±2.43**	**95.48±1.74**
Transformer	Feature Sequence	93.67±0.25	87.02±0.59	<u>85.19±0.56</u>	<u>84.92±0.45</u>	76.90±0.57	<u>74.61±2.35</u>	<u>63.85±0.90</u>	<u>64.89±1.19</u>	<u>93.09±0.30</u>	<u>83.57±0.39</u>	79.60±0.41	80.77±0.44
	Gray-scale Image	<u>95.73±0.34</u>	<u>89.21±1.63</u>	81.19±0.55	82.91±0.66	<u>87.53±0.66</u>	74.56±1.36	60.75±0.52	61.51±1.05	91.04±0.51	83.42±0.54	<u>79.70±0.79</u>	<u>80.83±0.75</u>
	Packet-Temporal Matrix	**98.40±0.20**	**89.75±1.65**	**91.74±1.68**	**90.41±1.50**	**92.53±0.28**	**89.50±0.55**	**84.47±0.46**	**86.65±0.41**	**98.96±0.09**	**89.50±1.41**	**83.15±0.77**	**84.18±1.10**

V. EXPERIMENTAL RESULTS AND ANALYSIS

To better evaluate the performance of the CL-ViME model, we pose the following four research questions (RQs):

RQ1: Is the proposed packet-temporal matrix representation superior to traditional feature representation methods?

RQ2: Can the proposed Vertical ViT-MoE model extract more effective information, thereby improving classification performance?

RQ3: Can the self-supervised contrastive learning pre-training strategy enhance the model's classification and generalization capabilities?

RQ4: Can the CL-ViME model outperform other state-of-the-art models?

To reduce experimental randomness, each experiment is repeated 30 times. Values before the "±" symbol indicate the mean, while those after indicate the standard deviation. In the experimental results, the best results are highlighted in bold and the second-best results are marked with underline.

A. Answer to RQ1

To answer RQ1, we select five classic models (CNN, VGG, RNN, LSTM, and Transformer) in a supervised learning setting. Experiments are conducted using three representation methods: feature sequence, gray-scale image, and packet-temporal matrix. Detailed results are shown in Table II.

As shown in Table II, the proposed packet-temporal matrix achieves the highest performance in most cases. This is mainly due to its structured fusion of dual-granularity information, combining fine-grained packet-level headers with macroscopic flow-level temporal patterns. Such a representation enriches feature information and enhances class separability, improving classification efficiency. On the ISCX-VPN dataset, it outperforms feature sequence and gray-scale image representations by 7.02% and 4.36% in accuracy, and 5.87% and 3.44% in macro-precision, respectively. On the CICIOT2023 dataset, the performance gain over gray-scale images is slightly reduced for some models, likely because IoT traffic exhibits simpler characteristics that are visually captured by less-structured representations. On other datasets, gray-scale images consistently underperform compared to the packet-temporal matrix. Overall, the proposed representation stores richer dual-granularity features, enhancing the performance and stability of supervised encrypted traffic classification. Additionally, sequence-based models like Transformer excel in these tasks, confirming their suitability as backbones for subsequent model design and optimization.

TABLE III
CLASSIFICATION PERFORMANCE OF DIFFERENT MODELS (%)

Dataset	Metric	ResNet18	AutoEncoder	GNN	Vit	Vertical ViT-MoE
ISCX-VPN	Acc	93.63±0.18	95.27±1.11	95.54±0.01	<u>97.64±0.69</u>	**99.46±0.06**
	Ma-Pre	74.87±0.59	78.69±3.25	79.23±0.05	<u>87.12±2.76</u>	**97.80±0.66**
	Ma-Rec	91.59±0.23	93.52±2.20	93.04±0.08	<u>96.25±0.77</u>	**98.09±0.40**
	Ma-F1	79.66±0.69	83.20±3.17	83.58±0.06	<u>90.65±2.26</u>	**97.94±0.43**
CICIOT2023	Acc	81.55±0.46	88.67±0.57	80.26±0.03	<u>91.74±0.31</u>	**94.87±0.13**
	Ma-Pre	69.25±0.52	78.88±0.78	67.63±0.02	<u>84.71±0.85</u>	**86.96±0.51**
	Ma-Rec	78.70±0.45	<u>85.92±1.21</u>	76.79±0.03	84.71±0.85	**90.83±0.46**
	Ma-F1	71.73±0.52	81.09±0.77	70.16±0.02	<u>86.61±0.63</u>	**88.62±0.39**
CTU-Malware	Acc	87.73±1.25	<u>97.76±0.23</u>	88.68±0.01	97.71±0.27	**98.94±0.29**
	Ma-Pre	66.31±1.14	86.32±2.41	67.90±0.01	<u>86.87±1.50</u>	**95.58±0.73**
	Ma-Rec	89.34±0.69	82.06±1.20	87.17±0.01	<u>92.14±1.49</u>	**96.82±0.80**
	Ma-F1	70.46±1.09	83.33±1.46	71.87±0.01	<u>88.67±1.22</u>	**96.02±0.48**

B. Answer to RQ2

To answer RQ2, we compare the proposed Vertical ViT-MoE model with ResNet18, AutoEncoder, GNN and ViT. All models are trained under identical self-supervised learning conditions. Experimental results are presented in Table III.

As shown in Table III, the proposed Vertical ViT-MoE outperforms the other four models across all evaluation metrics. On the CTU-Malware dataset, it achieves a macro-F1 of 96.02%, exceeding the best baseline (ViT, 88.67%) by 7.35%. This gain arises because the isotropic square patching in standard ViT disrupts the semantic integrity of highly structured traffic matrices. In contrast, Vertical ViT-MoE uses a vertical patching strategy aligned with the anisotropic nature of the data, preserving and extracting distinct packet-level and flow-level features. On the ISCX-VPN dataset, Vertical ViT-MoE reaches an accuracy of 99.46%, outperforming best baseline of ViT (97.64%) by 1.82%. Although standard ViT captures spatial correlations, its uniform segmentation is suboptimal for encrypted traffic, where semantics are organized by rows and columns. The vertical segmentation combined with the MoE architecture enables the model to learn specialized and robust representations, making it more suitable for encrypted traffic classification. Overall, Vertical ViT-MoE overcomes the limitations of standard vision models for structured encrypted traffic. By respecting the intrinsic structure of traffic, it captures both fine-grained packet-level headers and broader temporal-flow dynamics, resulting in improved performance and stability.

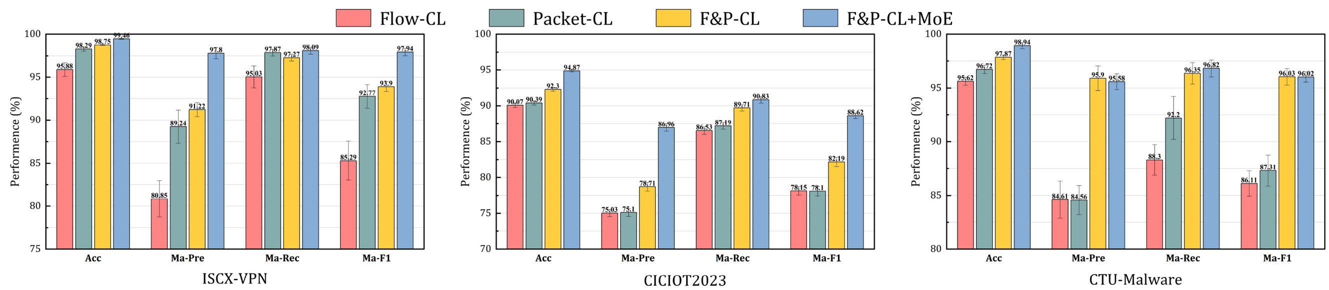


Fig. 3. Ablation experiments.

TABLE IV
CLASSIFICATION PERFORMANCE OF DIFFERENT CONTRASTIVE LEARNING MODELS (%)

Dataset	ISCX-VPN				CICIOT2023				CTU-Malware			
Metric	Acc	Ma-Pre	Ma-Rec	Ma-F1	Acc	Ma-Pre	Ma-Rec	Ma-F1	Acc	Ma-Pre	Ma-Rec	Ma-F1
MoCo [27]	89.43±0.14	69.99±0.82	70.06±1.11	69.97±0.41	85.58±0.11	75.54±0.20	76.65±0.33	76.06±0.24	93.92±0.09	77.23±1.23	77.96±0.49	77.38±0.71
SimCLR [28]	93.51±0.11	76.98±0.67	76.64±0.66	76.77±0.52	86.75±0.17	78.01±0.34	78.17±0.15	78.08±0.22	94.19±0.18	79.98±0.65	80.31±0.33	79.95±0.37
BYOL [35]	98.45±0.07	90.32±0.45	96.25±0.42	92.97±0.32	88.64±0.25	78.63±0.44	86.90±0.29	81.70±0.46	92.50±0.53	74.65±1.29	89.75±1.27	79.25±1.14
CL-ViME	**99.46±0.06**	**97.80±0.66**	**98.09±0.40**	**97.94±0.43**	**94.87±0.13**	**86.96±0.51**	**90.83±0.46**	**88.62±0.39**	**98.94±0.29**	**95.58±0.73**	**96.82±0.80**	**96.02±0.48**

C. Answer to RQ3

To answer RQ3, we select three representative self-supervised models of MoCo [27], SimCLR [28] and BYOL [35] as baselines, the experimental results are shown in Table IV. In addition, we also perform ablation experiments to verify the role of each component. Experiments are conducted under the following conditions: (1) with the MoE projector removed, including only flow-level contrastive learning (Flow-CL), only packet-level contrastive learning (Packet-CL), or dual-granularity contrastive learning (F&P-CL); and (2) with the MoE projector applied to dual-granularity contrastive learning (F&P-CL+MoE). All comparisons use identical self-supervised contrastive learning settings, with Vertical ViT-MoE as the backbone to ensure fairness. The results are shown in Fig. 3.

As shown in Table IV, the complete proposed model achieves optimal performance on all three encrypted traffic datasets. It significantly outperforms the baselines MoCo, SimCLR, and BYOL across all evaluation metrics. On the CICIOT2023 dataset, the macro-F1 of the proposed model reaches 88.62%, exceeding the best baseline (BYOL, 81.70%). This demonstrates the overall advantage of dual-granularity contrastive learning with the MoE projector.

Fig. 3 highlights the importance of each component. The following conclusions can be drawn: (1) Necessity of dual-granularity contrastive learning: On ISCX-VPN and CTU-Malware datasets, packet-level contrastive learning outperforms flow-level learning, while both achieve similar accuracy on CICIOT2023. Combining the two granularities consistently surpasses single-granularity models, validating their complementarity. For instance, on CICIOT2023, the dual-granularity model achieves 92.30% accuracy, higher than flow-level (90.07%) or packet-level (90.39%). (2) Effectiveness of MoE projector: Introducing the MoE projector further improves performance. On ISCX-VPN, accuracy rises from 98.75% to 99.46% and macro-F1 from 93.90% to 97.94%. This confirms that the MoE projector and auxiliary loss L_{moe} effectively map heterogeneous packet- and flow-level features into a unified latent space and learn more discriminative semantic features through dynamic expert routing.

Overall, experimental results verify that the proposed self-supervised pretraining model enhances model generalization. Its success relies on two aspects: (1) collaborative use of packet- and flow-level features via structured data augmentation (matrix transposition), and (2) efficient integration and refinement of heterogeneous features through the MoE projector and auxiliary loss. These mechanisms together enable the model to learn rich and discriminative representations from unlabeled encrypted traffic.

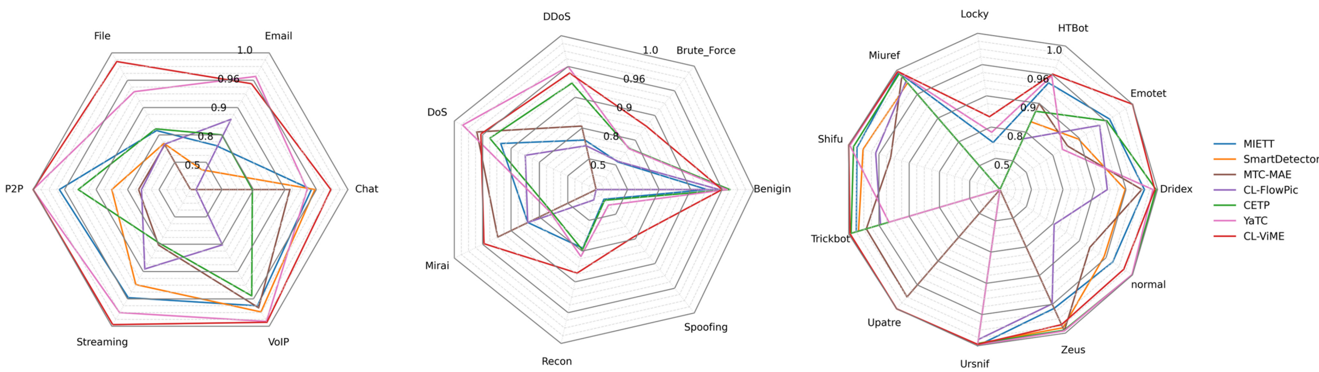
D. Answer to RQ4

To answer RQ4, we compare the proposed CL-ViME model with seven state-of-the-art models.The detailed results are presented in Table V.

As shown in Table V, the proposed CL-ViME model achieves both high accuracy and remarkable stability compared to other traffic classification models. On the ISCX-VPN and CICIOT2023 datasets, it ranks first on all four metrics, outperforming competitors by a substantial margin. On CICIOT2023, its macro-F1 reaches 88.62%, 8.6% higher than the runner-up YaTC, demonstrating superior discriminative power. For YaTC, the full-model fine-tuning used in downstream classification can induce catastrophic forgetting in the backbone, weakening the pre-training benefits. In addition, its masked autoencoder is optimized for token reconstruction, which biases the learned representations toward structural fidelity rather than discriminative separability. This design limits its effectiveness in classification tasks. On the CTU-Malware dataset, CL-ViME shows robustness to class imbalance. Although CETP slightly surpasses it in raw accuracy (99.04% vs. 98.94%), its macro-averaged metrics lag far behind. This "high accuracy, low macro" profile indicates overfitting to majority classes. The macro metrics weight each class equally, providing a more realistic measure of performance. Here, CL-ViME achieves a macro-F1 of 96.02%, exceeding

TABLE V
PERFORMANCE OF DIFFERENT ENCRYPTED TRAFFIC CLASSIFICATION MODELS (%)

Dataset	ISCX-VPN				CICIOT2023				CTU-Malware			
Metric	Acc	Ma-Pre	Ma-Rec	Ma-F1	Acc	Ma-Pre	Ma-Rec	Ma-F1	Acc	Ma-Pre	Ma-Rec	Ma-F1
MIETT [19]	94.81±0.15	93.18±0.44	89.26±0.31	90.90±0.21	83.61±0.18	75.86±0.55	69.92±0.35	71.83±0.38	97.42±0.08	87.74±0.49	85.39±0.39	86.31±0.38
SmartDetector [18]	94.30±0.32	87.32±1.48	80.44±1.36	81.83±1.60	89.54±0.11	81.08±1.55	67.71±0.49	69.71±0.48	97.65±0.10	82.50±4.17	81.46±3.97	81.94±4.06
MTC-MAE [15]	89.19±0.21	66.86±0.26	67.44±0.54	67.09±0.28	82.39±0.09	46.65±0.04	51.74±0.07	48.99±0.05	94.11±0.33	77.30±0.27	77.34±0.27	77.24±0.27
CL-FlowPic [33]	70.61±0.78	72.29±1.14	66.55±0.80	65.50±0.72	77.74±0.52	67.48±1.03	56.98±0.78	57.19±1.18	90.58±0.29	74.78±0.28	75.20±0.31	74.91±0.23
CETP [17]	90.14±0.14	85.04±0.81	84.73±0.84	84.61±0.92	91.54±0.29	76.65±0.45	74.62±1.03	74.63±0.74	**99.04±0.19**	80.72±0.20	80.13±0.40	80.41±0.29
YaTC [34]	97.19±0.09	97.37±0.07	96.90±0.07	97.11±0.07	91.92±0.03	83.86±0.09	77.54±0.15	80.02±0.09	98.72±0.01	89.15±0.33	86.12±0.06	87.51±0.12
CL-ViME	**99.46±0.06**	**97.80±0.66**	**98.09±0.40**	**97.94±0.43**	**94.87±0.13**	**86.96±0.51**	**90.83±0.46**	**88.62±0.39**	98.94±0.29	**95.58±0.73**	**96.82±0.80**	**96.02±0.48**



ISCX-VPN CICIOT2023 CTU-Malware

Fig. 4. Performance on each category of encrypted traffic.

CETP by over 15%, confirming its ability to maintain high precision across both dominant and rare classes.

Fig. 4 presents radar charts of classification performance for seven contrastive learning-based models. CL-ViME outperforms rivals in most categories and achieves comparable performance in the remaining ones. While other models occasionally misclassify specific classes completely, CL-ViME does not fail catastrophically. This robustness is attributed to three key designs: (1) structured data augmentation (row-column transposition and random masking) generates high-quality positive sample pairs while preserving semantic structures; (2) the Vertical ViT-MoE architecture efficiently extracts anisotropic features of encrypted traffic; (3) the combination of hybrid bidirectional contrastive loss and MoE auxiliary loss provides a comprehensive optimization target. Overall, CL-ViME demonstrates excellent classification performance, strong stability, and robustness across diverse encrypted traffic datasets, especially under class imbalance.

E. Discussion

Existing contrastive learning-based encrypted traffic classification models suffer from three main limitations. (1) They slice encrypted traffic into fixed-length byte sequences or 2-D matrices, discarding fine-grained packet-header semantics and severing macroscopic temporal structures. (2) Directly transferring vision models distorts the anisotropic structure of traffic matrices. (3) They exploit only a single granularity, failing to leverage the complementarity of packet- and flow-level information. To address these issues, we propose CL-ViME. It introduces a packet-temporal matrix that fuses packet-level headers with flow-level temporal patterns. The Vertical ViT-MoE employs vertical patching and dynamic expert routing to respect the traffic structure. A dual-granularity contrastive learning pretraining scheme aligns packet- and flow-level representations without labels, substantially boosting robustness and generalization.

CL-ViME is evaluated on three encrypted traffic datasets against advanced self-supervised and supervised models. It achieves accuracies of 99.46%, 94.87% and 98.94%, outperforming all baselines. Even under severe class imbalance, it maintains high precision on minority classes, demonstrating its effectiveness in feature representation, model architecture and pretraining design.

Despite its strengths, CL-ViME has following limitations: (1) Current design is computationally expensive due to the large embedding dimension (120) and the numerous vertical patches (e.g., 30 for a 60×60 input). Future work will explore replacing PatchEmbed with a CNN-based module that performs spatial downscaling, thereby reducing the sequence length and embedding dimension while preserving essential traffic structures. (2) Dynamic network environment may cause concept drift, significantly degrading the performance of CL-ViME model. To address this challenge, we can incorporate incremental or continual learning mechanisms, allowing the model to adapt to evolving traffic distributions without catastrophic forgetting.

F. Threats to Validity

This section discusses potential threats to the internal and external validity of the CL-ViME model during experimental evaluation.

Internal validity. First, there are potential threats in reproducing baselines. Some models, such as CETP, MIETT, SmartDetector and CL-FlowPic, do not provide official source codes. We reproduced them based on the descriptions in the papers. Although we follow the algorithm descriptions and experimental settings, complete consistency with the original implementations cannot be guaranteed. Second, hyperparameter optimization may introduce bias. Although the same search space and optimization strategy are used for all models, different models have varying sensitivity to hyperparameters. The CL-ViME model has been thoroughly tuned, but some baselines may not reach their optimal performance.

External validity. Threats to external validity concern the generalizability of the results. The three datasets used (ISCX-VPN, CICIoT2023, CTU-Malware) cover diverse network environments and encryption protocols. However, they do not represent all possible encrypted traffic scenarios. Therefore, the results may not fully capture nuances of emerging encryption techniques or novel attack vectors.

VI. Conclusion and Future Work

To address the challenges of incomplete feature representation and limited labeled data in encrypted traffic classification, this paper proposes CL-ViME, a self-supervised encrypted traffic classification model built upon three key technical components. First, we introduce a packet-temporal matrix that fuses fine-grained packet headers with macroscopic flow-level temporal patterns, forming a more comprehensive and informative representation of encrypted traffic. Second, we design a Vertical ViT-MoE architecture tailored to traffic structure. It uses vertical patching to preserve packet sequencing and employs dynamic expert routing to select the most suitable expert subnetworks, thereby improving representational quality and computational efficiency. Third, we develop a dual-granularity contrastive pretraining scheme that removes the need for manual labels. It constructs complementary packet-level and flow-level views of the same traffic sample and aligns their representations, enabling the model to learn essential and invariant features in an unsupervised manner. These components jointly enhance the robustness and generalization capability of CL-ViME.

Extensive experiments validate the effectiveness and superiority of the proposed CL-ViME model. Results on three public encrypted traffic datasets show that CL-ViME consistently surpasses state-of-the-art models. In particular, it maintains strong robustness and generalization under class imbalance in real-world malware traffic, highlighting its potential for deployment in complex network environments. However, the CL-ViME model suffers from high computational cost and limited robustness to distribution shifts introduced by emerging encryption protocols. Future work should investigate lightweight architectures to enhance efficiency and explore incremental or continual learning mechanisms to improve model's adaptability to evolving traffic distributions.

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